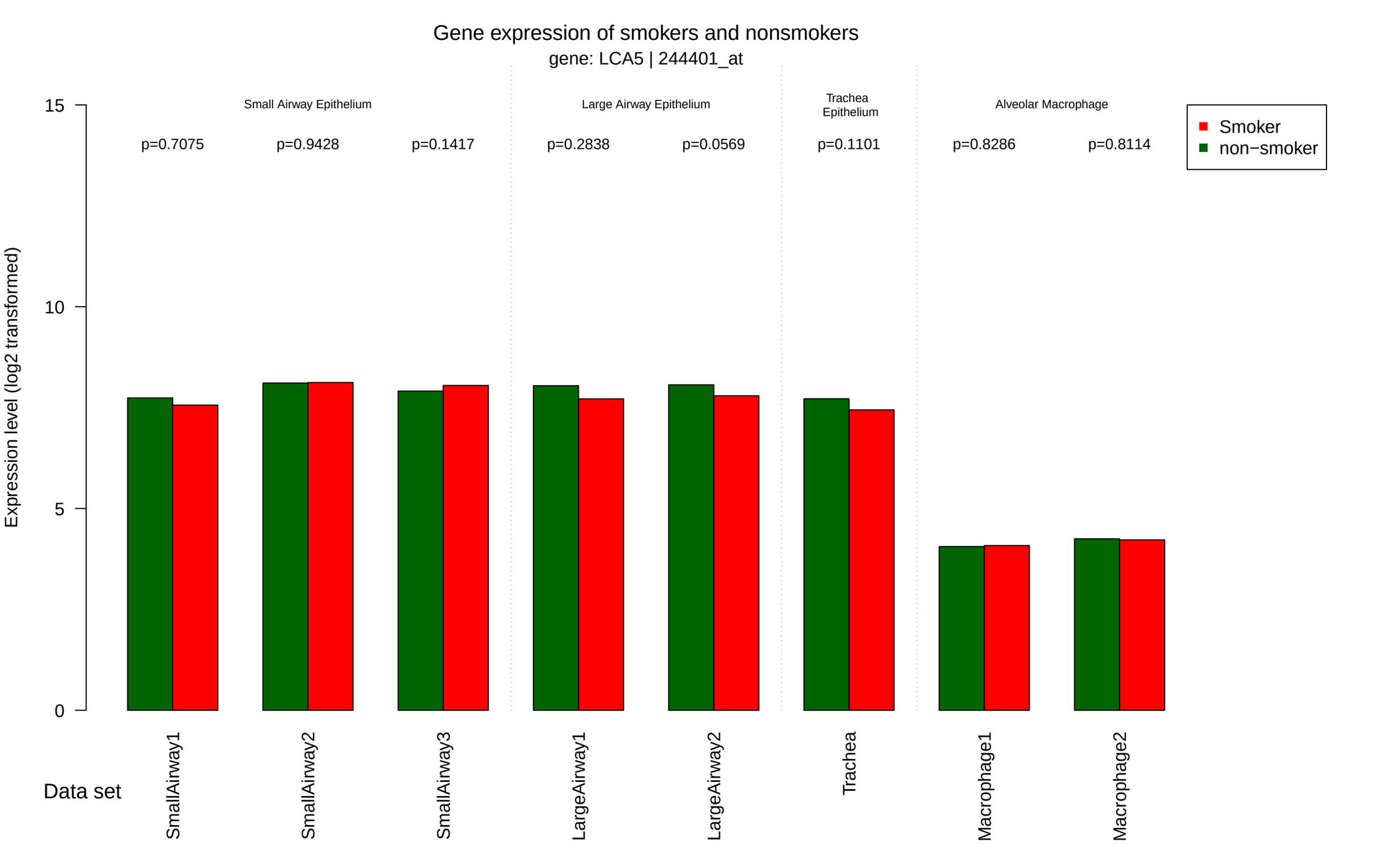
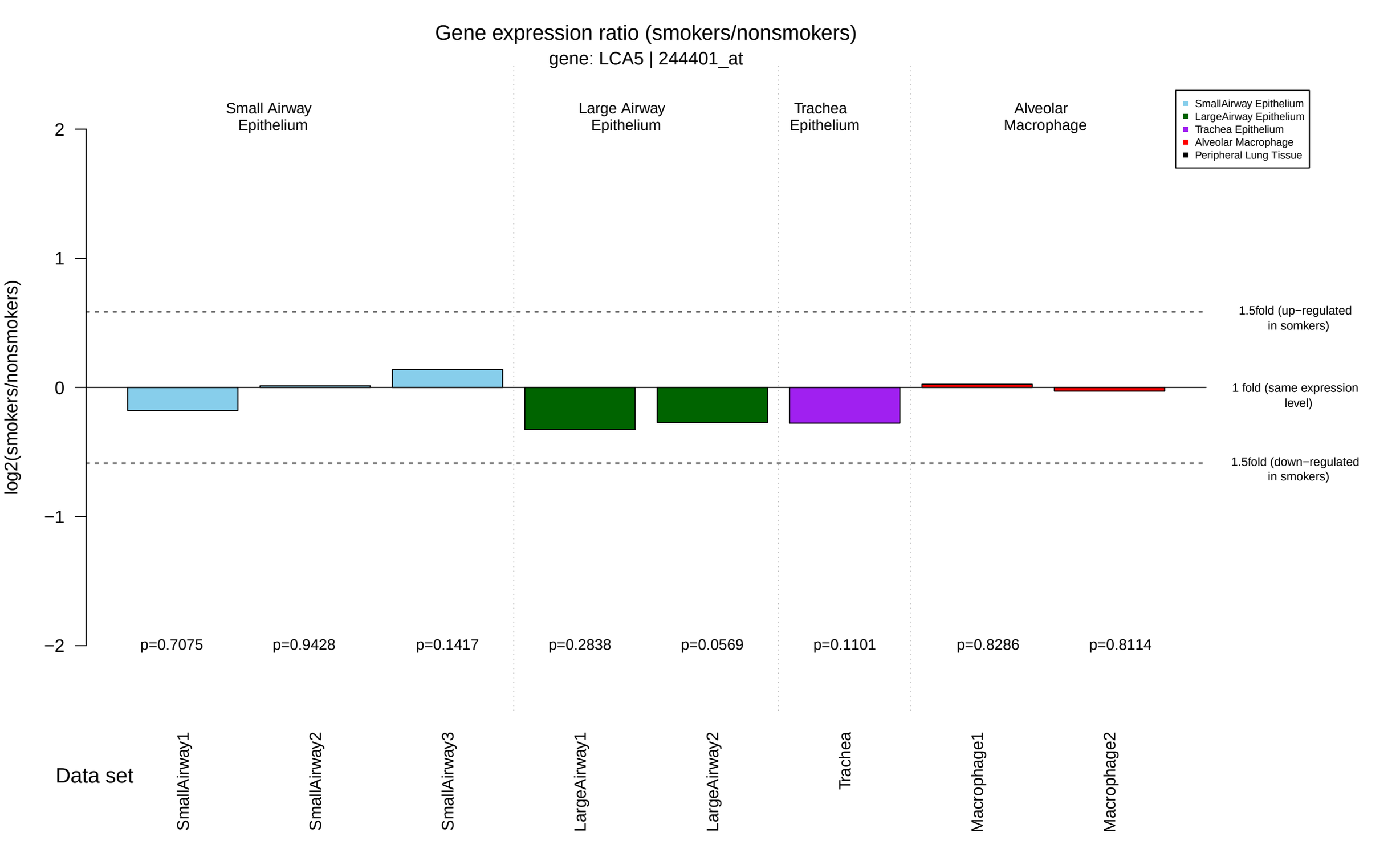
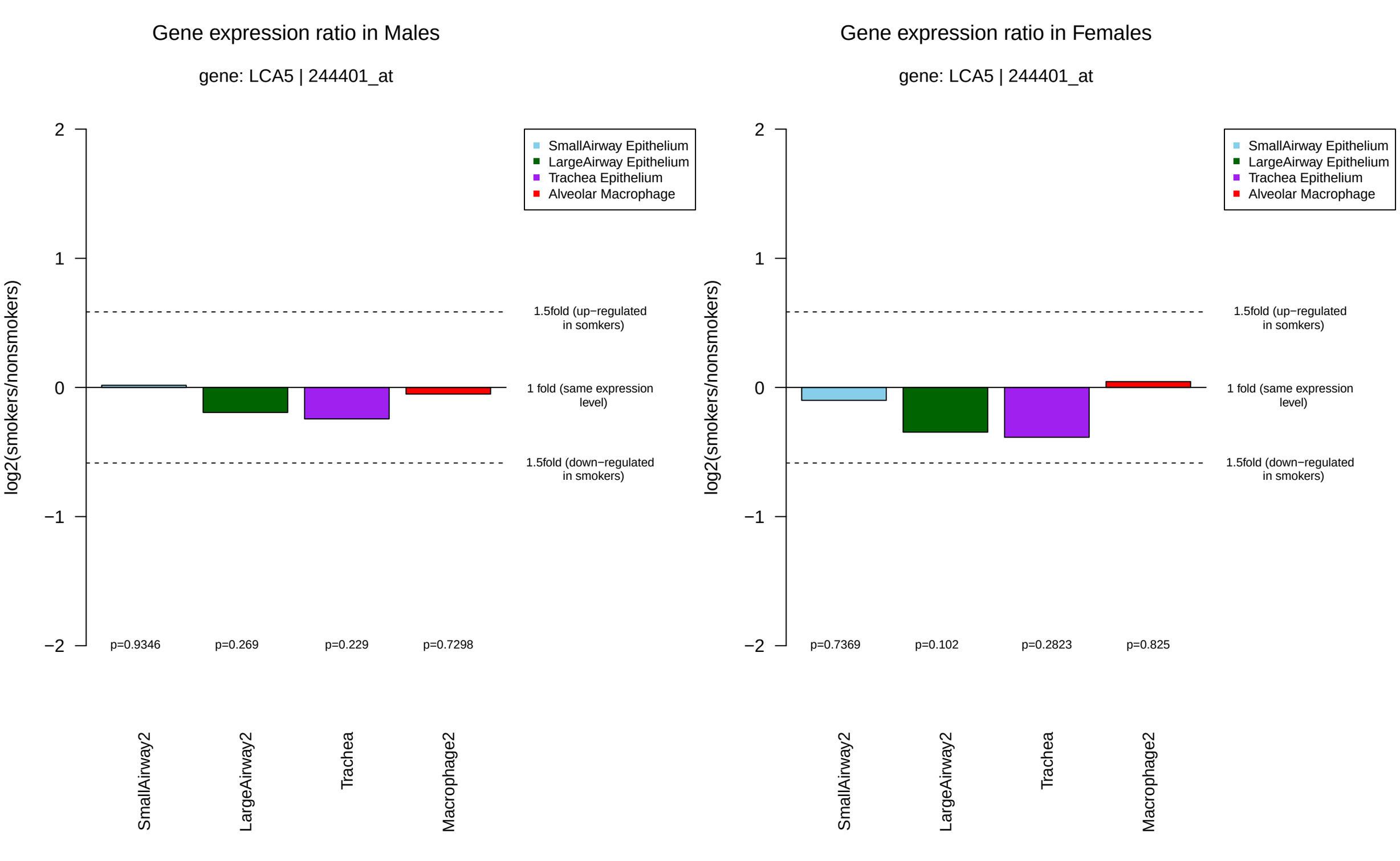
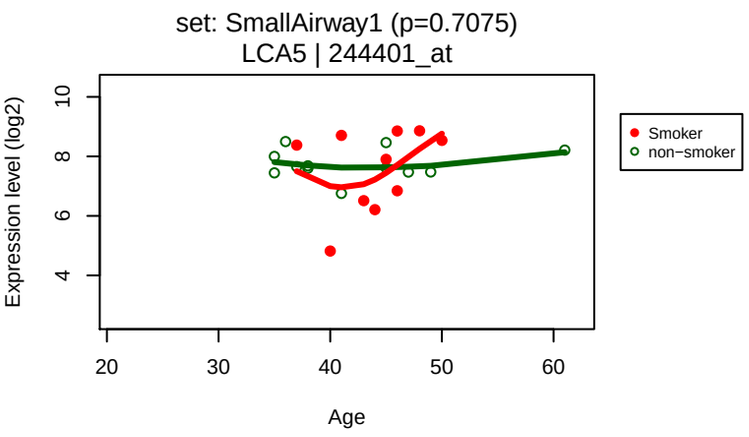
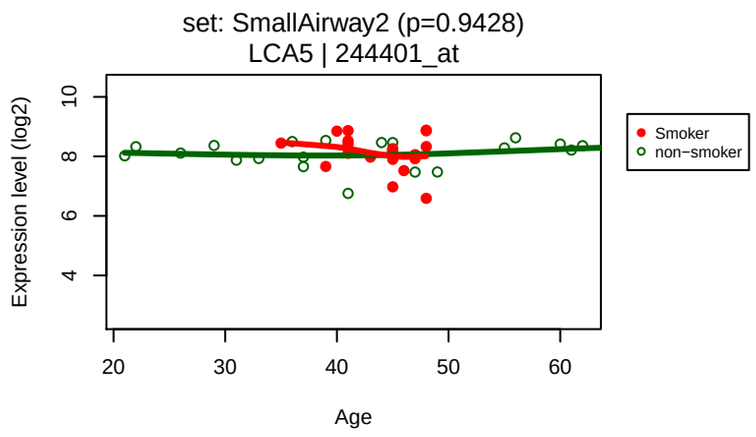
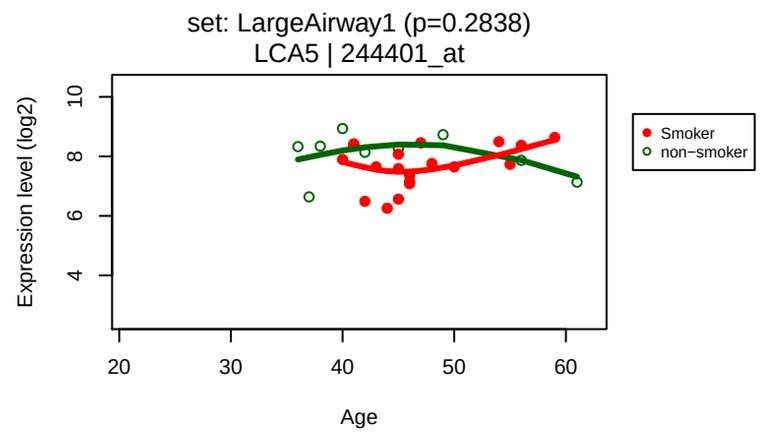

Gene expression of smokers and nonsmokers
gene: LCA5 | 244401_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7075
p=0.9428
p=0.1417
p=0.2838
p=0.0569
p=0.1101
p=0.8286
p=0.8114
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

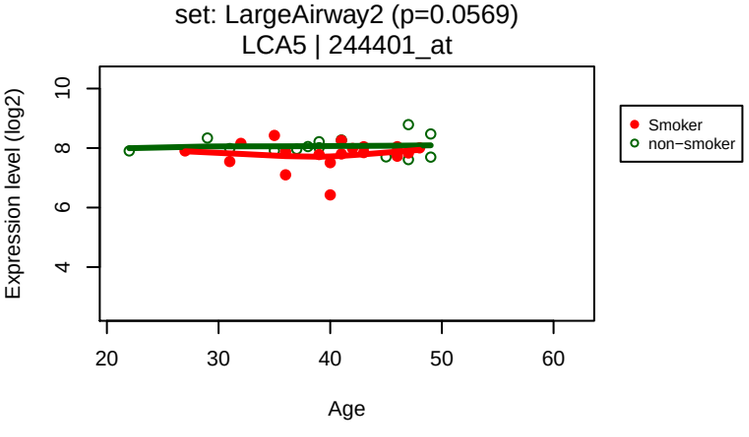

Gene expression ratio (smokers/nonsmokers)
gene: LCA5 | 244401_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7075
p=0.9428
p=0.1417
p=0.2838
p=0.0569
p=0.1101
p=0.8286
p=0.8114
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

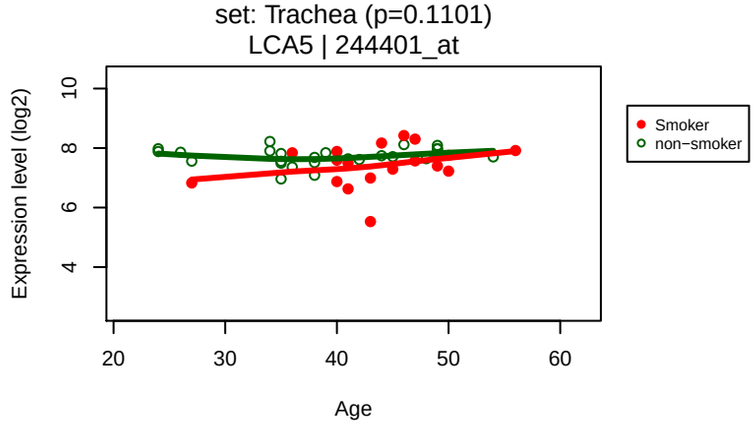

Gene expression ratio in Males
gene: LCA5 | 244401_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9346
p=0.269
p=0.229
p=0.7298
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LCA5 | 244401_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7369
p=0.102
p=0.2823
p=0.825
SmallAirway2
LargeAirway2
Trachea
Macrophage2

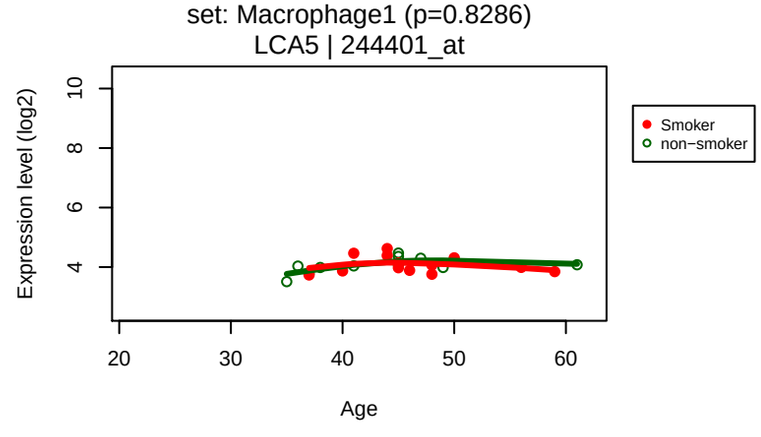

set: SmallAirway1 (p=0.7075)
LCA5 | 244401_at
Expression level (log2)
Age
Smoker
non−smoker

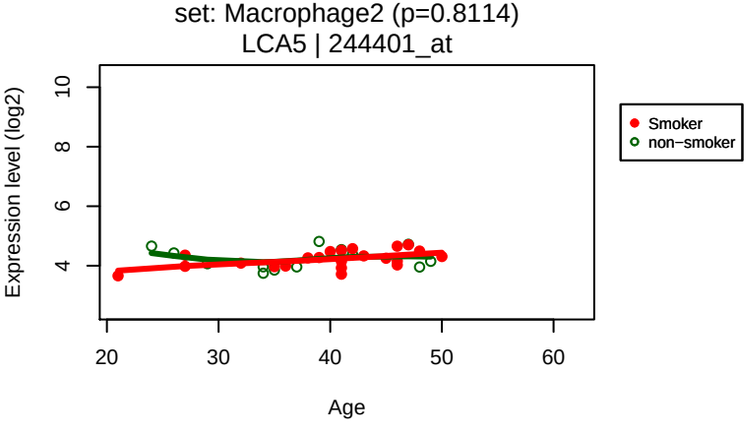

set: SmallAirway2 (p=0.9428)
LCA5 | 244401_at
Expression level (log2)
Age
Smoker
non−smoker

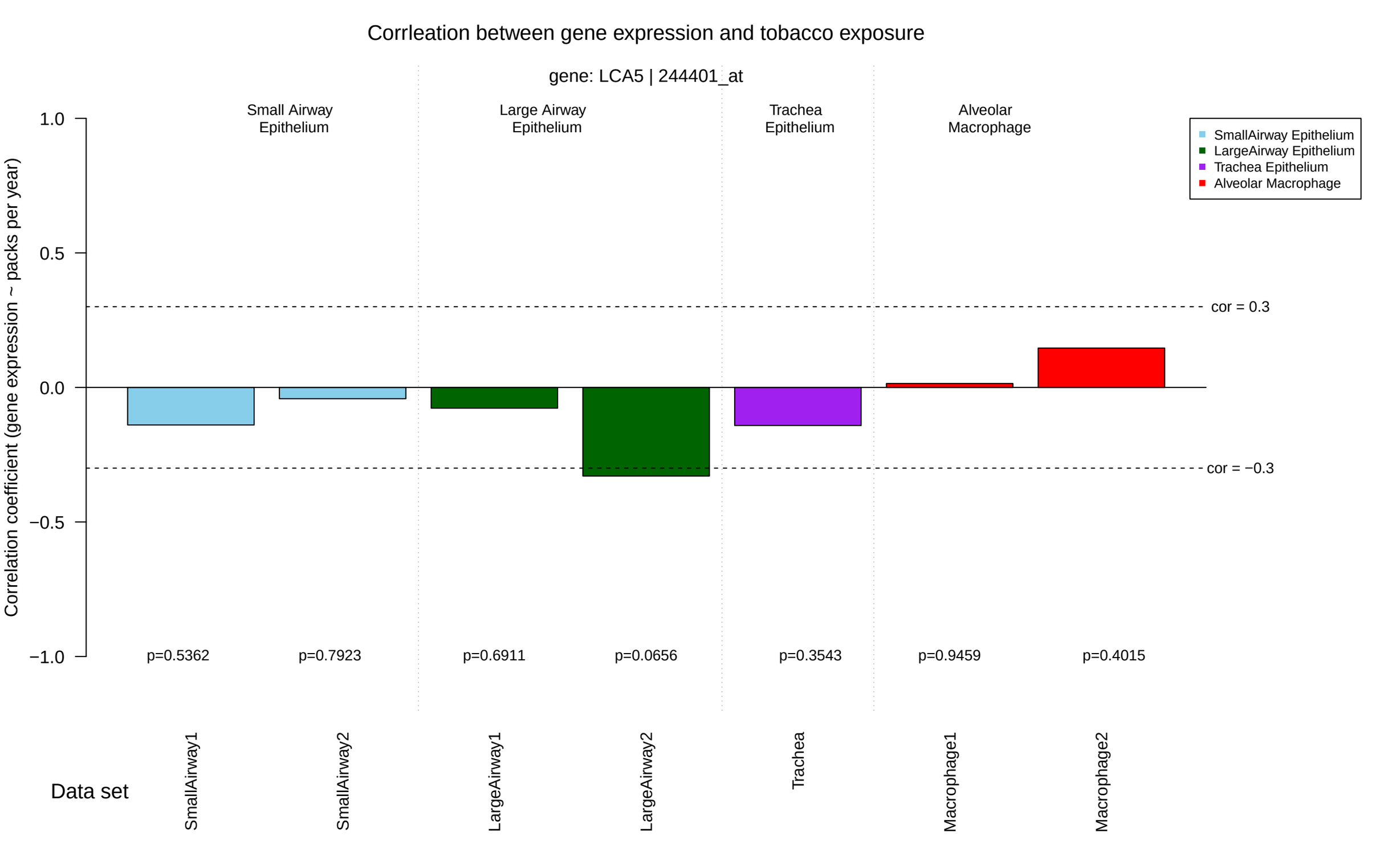

set: LargeAirway1 (p=0.2838)
LCA5 | 244401_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0569)
LCA5 | 244401_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1101)
LCA5 | 244401_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8286)
LCA5 | 244401_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8114)
LCA5 | 244401_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LCA5 | 244401_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5362
p=0.7923
p=0.6911
p=0.0656
p=0.3543
p=0.9459
p=0.4015
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2